

Vimal S

Data & GenAI Engineer | Aspiring MLOps Engineer

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Data & GenAI Engineer with 7+ years of experience designing production-scale data platforms and AI solutions on AWS, Databricks and PySpark. Deliver measurable impact by building robust ETL/ELT pipelines, data models, and GenAI integrations (Bedrock, Strands Agent Framework) that automate insight delivery and decision-making. Currently expanding MLOps expertise to accelerate model deployment, monitoring, and reliability.

Skills

- **Data & Cloud Platforms:** AWS (EMR, Glue, S3, Redshift, Lambda, DynamoDB, Step Functions, Bedrock, SageMaker), Databricks, Snowflake, Kafka, Power BI
- **Programming & Frameworks:** Python, SQL, PySpark, Oracle PL/SQL, Pandas, Boto3, Scikit-learn, TensorFlow.
- **GenAI :** Prompt engineering, Amazon Bedrock AgentCore, Strands, MLflow (GenAI), Deepeval
- **Data Engineering & Ops:** ETL/ELT design, Data Lake & Data Warehouse modeling, Airflow, ActiveBatch, Terraform, CI/CD (GitHub Actions), Docker
- **Analytics & Visualization:** Power BI, KPI dashboarding (retail use cases)

Work Experience

EPAM Systems – Hyderabad, Telangana

Data & GenAI Engineer (May 2025 - Current)

- Delivered an end-to-end GenAI customer-support chat assistant for a telecom contact center, orchestrating Bedrock agentcore , DynamoDB tool-calling and RAG resulting in faster and efficient customer handling.
- Designed and productionized a multi-agent GenAI chat assistant on Strands, instrumenting MLflow + RAGAS evaluation to track accuracy/latency, and automating a validation pipeline that streams metrics into CloudWatch and triggers SNS alerts to stakeholders whenever performance slips below SLA thresholds.
- Orchestrated the end-to-end churn pipeline using Step functions, Glue and Sagemaker across staging, enrichment, and consumption layers, giving SageMaker pipelines consistent inputs and keeping dashboards aligned with current customer behavior.
- Have implemented MLOps principles and tools as of Churn prediction ML model using Sagemaker features such as Feature store, Model Registry, Model Monitor, Sagemaker Clarify and Sagemaker Pipelines.

Cognizant Technology Solutions - Chennai, Tamil Nadu

Data Engineer (July 2021 - May 2025)

- Built a reusable PySpark ingestion and transformation framework that standardized retail KPI derivation, feeding the Power BI team with trusted, analytics-ready datasets
- Implemented PySpark optimization guardrails—partitioning, bucketing, executor sizing, and caching—that kept nightly pipelines stable, ensured KPI deliveries hit the Power BI refresh window, and maintained tight control over compute spend.
- Collaborated with data analysts on Power BI dashboard initiatives—modeling curated datasets and advising on layout best practices-demonstrating working knowledge of dashboard design and development.

Cognizant Technology Solutions - Chennai, Tamil Nadu

PL SQL developer (June 2019 - June 2021)

- Enhanced core insurance modules—commissions, fund exchange, riders, and claims—by refactoring PL/SQL packages and rapidly implementing the frequent business-rule changes required to keep production systems aligned with evolving product and regulatory updates.
- Built automated PL/SQL-driven reporting and data-transfer routines that consistently delivered finance-ready extracts on schedule, giving operations timely, validated data for analysis and record keeping.

Education

PSG College of Technology - Coimbatore, Tamil Nadu

July 2015 – May 2019

B.E in Electronics and Communication Engineering | 7.88 CGPA

Certifications

- AWS certified Data Engineer Associate (DEA- C01)
- AWS certified Machine Learning Engineer Associate (MLA – C01)
- Databricks Certified Data Engineer Associate
- Databricks Certified Generative AI Engineer Associate
- Snowpro core certification
- Microsoft Certified: Azure Fundamentals (AZ-900)

Projects

Customer 360 Analytics Solution: C360 is an analytics platform for a telecom client that consolidates customer, transaction, and product data. I built ETL pipelines to integrate multiple data sources, applied data quality rules, and developed a Gold layer dataset. This enabled SageMaker churn prediction and delivered actionable insights through Quicksight dashboards.

Customer Care QnA chatbot Assistant: This project aims to build an intelligent assistant for customer care agents, providing quick access to product details, solutions for issues, and customer information like billing & plan details, identifies high-risk churn customers and suggests targeted actions to improve retention and customer satisfaction.

Retail Data Analytics Platform : A comprehensive analytics solution was developed to process and transform large volumes of retail data using scalable ETL pipelines on AWS EMR. The resulting data was stored in Amazon Athena and Redshift, enabling data analysts to build business visualizations in Quicksight and drive informed decisions.

Unified Analytics platform for Insurance : A centralized analytics solution developed to unify and curate data from numerous insurance applications using AWS and Databricks. The resulting "gold layer" and secure data marts established a single source of truth for business KPIs, enabling the analytics team with reliable, visualization-ready data to enhance reporting and drive data-driven decisions.

Insurance Application for Annuity and Life Products : Built for the insurance sector, this application delivers individual annuity and life insurance offerings, managed through a PL/SQL admin system with modules covering Riders, Rate Exchange, Death Claims, Commissions, Annuitization, and other key functions.